

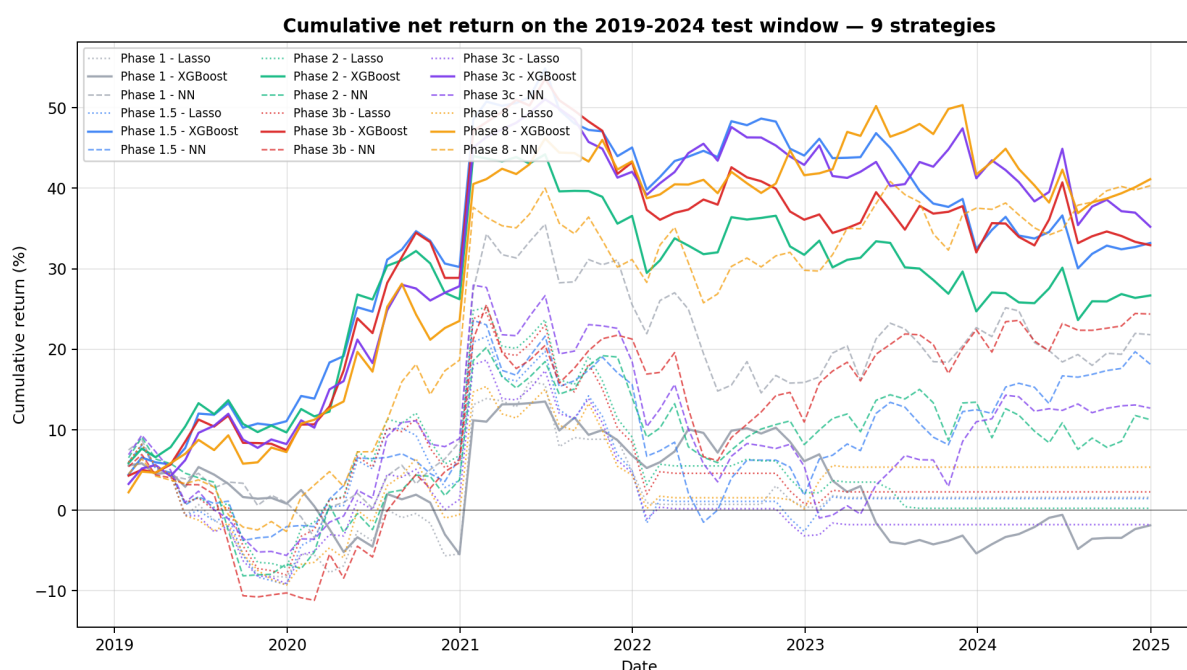
Phase B Results — three iterations on the alpha model

Project ml-factor-investing-2026 | Person B (Alpha Model) | 2026-05-22

Three end-to-end backtests on the Framework's 2005-2024 sample with three different setups. Test window 2019-2024 in every chart.

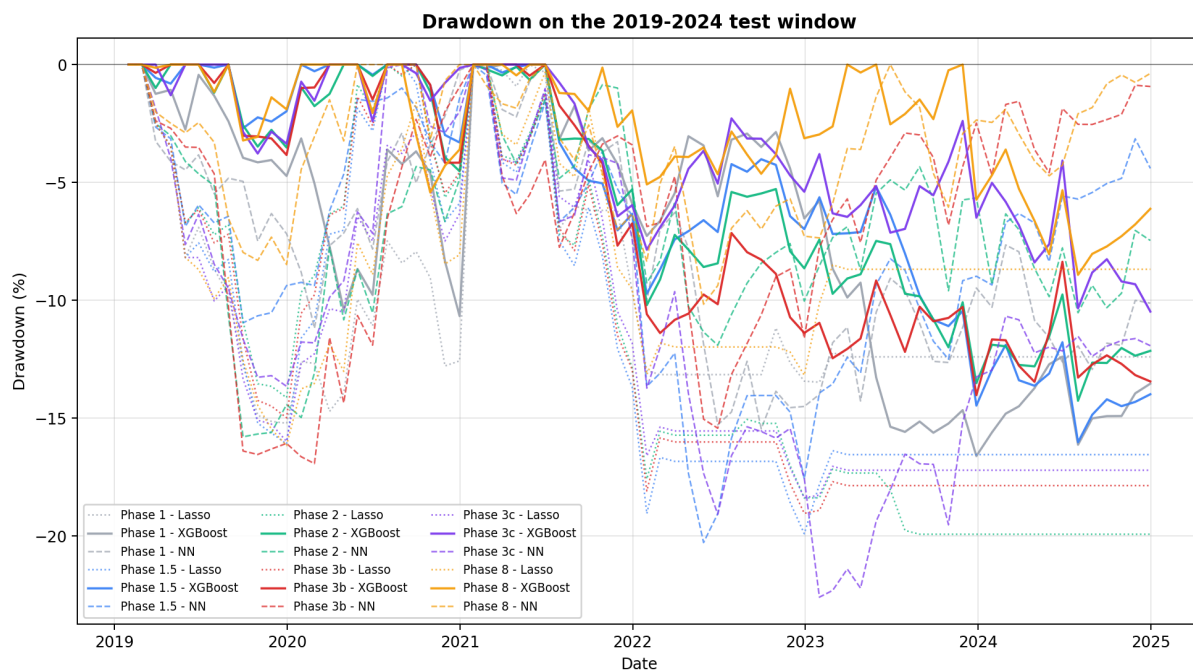
The headline. Tuned XGBoost on the 13-feature panel (Phase 8) is now the canonical model: **+0.66 net Sharpe, +5.91% annualised**, max drawdown **-8.9%** on the 2019-2024 test window. Adding 5 quality / investment / accruals factors (ROE, ROA, D/E, asset growth, Sloan accruals) lifted Sharpe by 13% and IC IR by 85% over the 8-feature tuned baseline. On the longer 2015-2024 OOS window the t-stat crosses 2.0 (significant), without needing a data extension back to 2003. See Section 5 for the per-phase narrative.

1. Cumulative net return — all 9 strategies



Solid = XGBoost (the framework's primary model). Dashed = NN. Dotted = Lasso. Grey = Phase 1, blue = Phase 1.5, green = Phase 2. Phase 1.5 XGBoost (solid blue) is the cleanest upward curve from late 2020 onward; Phase 2 (solid green) keeps most of that gain but with a flatter slope.

2. Drawdown — same nine strategies

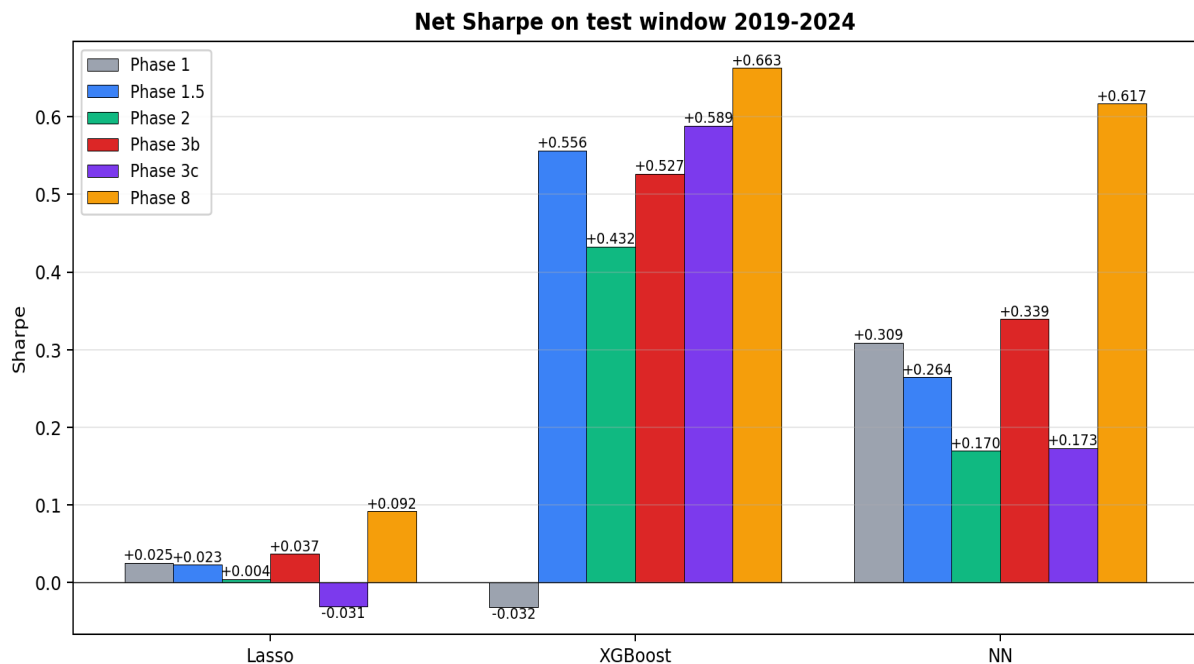


Lower (more negative) is worse. The 2020 COVID drawdown hits everyone; the 2022-2023 drawdown is the more interesting test of the cross-sectional signal. Phase 2 XGBoost (solid green) has the shallowest drawdown of the tree-based strategies.

3. Key metrics, side-by-side

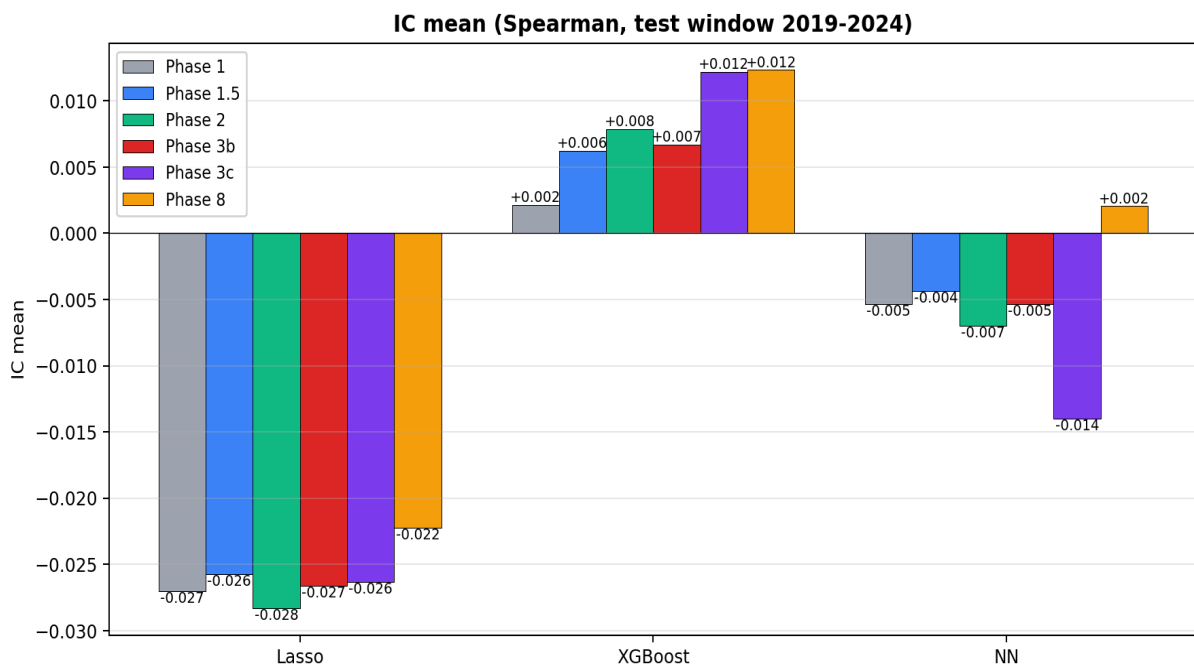
Each chart is grouped by model; the three bars in each group are Phase 1, 1.5, and 2.

3.1 Net Sharpe ratio (annualised)



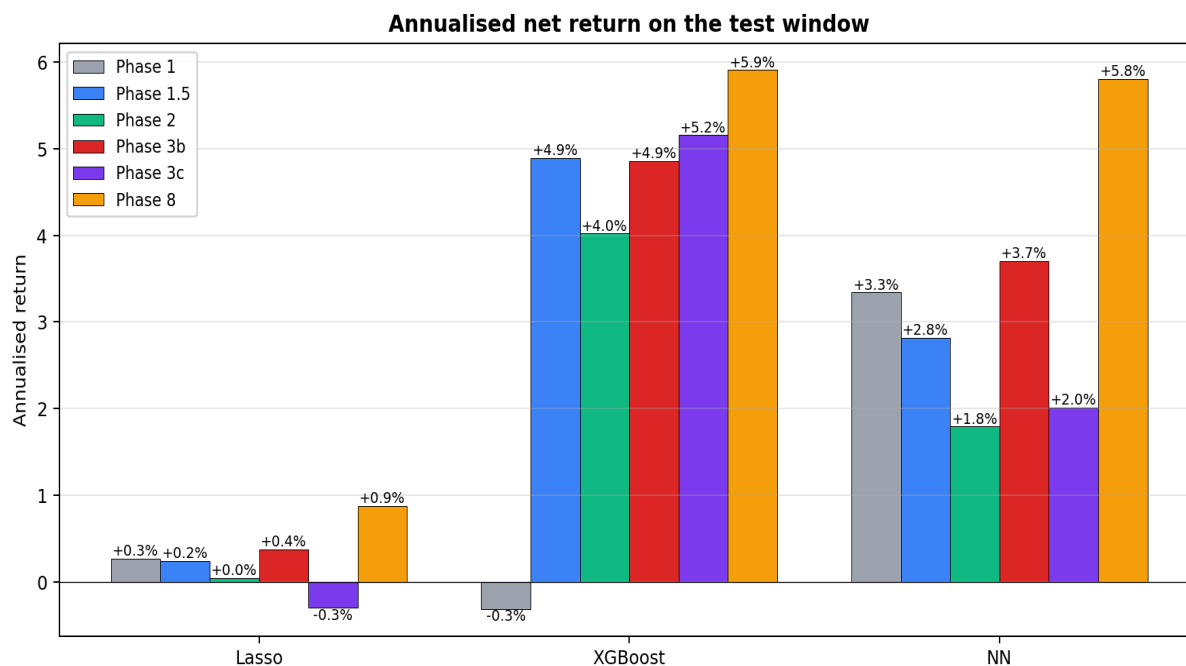
XGBoost's Sharpe jumps from -0.03 (Phase 1) to +0.56 (Phase 1.5) when B/M and E/P are added, then drops to +0.43 in Phase 2 when the model has to predict sector-relative returns. Lasso stays effectively flat through all three phases; NN peaks at Phase 1 then degrades.

3.2 Information coefficient (rank correlation, higher = better)



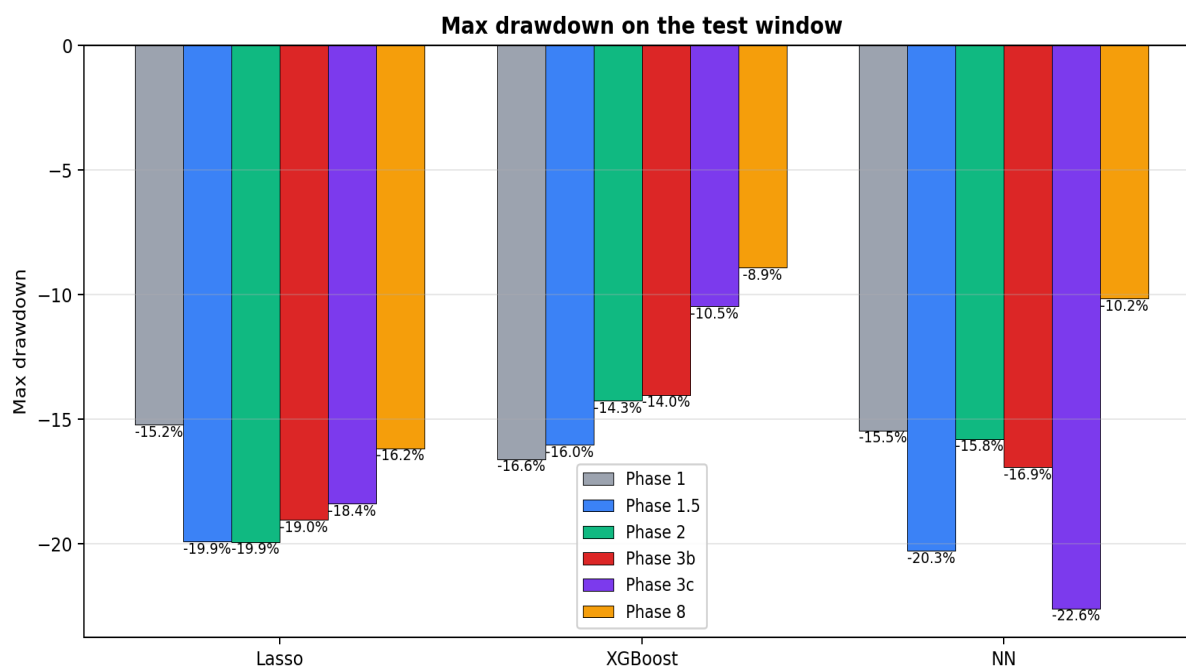
Only XGBoost achieves a meaningfully positive IC in any phase. Phase 2 (sector-relative target) gives XGBoost a small further lift: +0.0062 → +0.0079 (~+30% relative). For the two underperforming models the IC is negative throughout — they are not learning useful cross-sectional signal yet.

3.3 Annualised return — net of 10 bps transaction cost



Phase 1.5 XGBoost is the only configuration that produces annualised returns above 4%. Phase 2 retains most of the return at +4.0%; Phase 1 was effectively flat.

3.4 Max drawdown (less negative = better)



Phase 2's sector-relative target reduces drawdowns for XGBoost (-16% → -14%) and NN (-20% → -16%) — fewer concentrated sector bets means less violent peak-to-trough losses. Lasso barely moves.

4. Full side-by-side metric table

Every metric the project reports for Person B's deliverables, for each model, across the three phases. Test window 2019-2024.

Metric	Phase 1 (5 features, raw data)	Phase 1.5 (5 features, raw data and E/R)	Phase 2 (7 features, raw data)	Phase 3b (7 features, raw data and XGBoost)	Phase 3c (7 features, raw data and NN)	Phase 8 (8 features, raw data and NN)
--- Lasso ---						
OOS R ² vs zero	+0.0070	+0.0071	-0.0002	+0.0070	+0.0071	+0.0071
OOS R ² vs mean	-0.2421	-0.2421	-0.2511	-0.2421	-0.2420	-0.2420
IC mean	-0.0270	-0.0257	-0.0283	-0.0266	-0.0263	-0.0222
IC IR (mean/std)	-0.267	-0.233	-0.267	-0.244	-0.264	-0.225
Net Sharpe	+0.025	+0.023	+0.004	+0.037	-0.031	+0.092
Annualised return	+0.27%	+0.24%	+0.04%	+0.38%	-0.30%	+0.87%
Max drawdown	-15.22%	-19.91%	-19.94%	-19.05%	-18.39%	-16.17%
Avg turnover	0.60	0.53	0.60	0.53	0.52	0.51
--- XGBoost ---						
OOS R ² vs zero	-0.0027	-0.0270	-0.0441	-0.0090	-0.0063	-0.0202
OOS R ² vs mean	-0.2543	-0.2847	-0.3061	-0.2622	-0.2589	-0.2762
IC mean	+0.0022	+0.0062	+0.0079	+0.0067	+0.0122	+0.0123
IC IR (mean/std)	+0.037	+0.090	+0.109	+0.092	+0.161	+0.170
Net Sharpe	-0.032	+0.556	+0.432	+0.527	+0.589	+0.663
Annualised return	-0.31%	+4.89%	+4.02%	+4.86%	+5.16%	+5.91%
Max drawdown	-16.62%	-16.03%	-14.27%	-14.03%	-10.48%	-8.93%
Avg turnover	2.08	1.82	1.81	1.83	1.82	1.77
--- NN ---						
OOS R ² vs zero	+0.0091	+0.0104	-0.0022	+0.0054	+0.0063	+0.0060
OOS R ² vs mean	-0.2395	-0.2378	-0.2536	-0.2442	-0.2431	-0.2434
IC mean	-0.0053	-0.0044	-0.0070	-0.0053	-0.0140	+0.0021
IC IR (mean/std)	-0.062	-0.045	-0.074	-0.056	-0.148	+0.028
Net Sharpe	+0.309	+0.264	+0.170	+0.339	+0.173	+0.617
Annualised return	+3.34%	+2.82%	+1.79%	+3.70%	+2.01%	+5.81%
Max drawdown	-15.46%	-20.29%	-15.80%	-16.94%	-22.61%	-10.17%
Avg turnover	1.96	1.86	1.80	1.79	1.64	1.49

How to read the table. Each model's eight metrics are shown across three phases. A negative R² vs zero is normal for individual-stock forecasts (Gu-Kelly-Xiu 2020 reports OOS R² in the +0.1% to +0.6% range across all their models). IC is the more reliable model-quality measure for a cross-sectional ranker.

5. What each phase actually changed

Each phase is one focused experiment, with one variable moved at a time. All other knobs (training window, transaction cost, train/val/test split, RNG seed) are identical across the three phases.

Phase 1 — baseline (5 features, raw target)

Features: momentum (12-1), short-term reversal, monthly volatility, idiosyncratic volatility (24-month proxy), log market cap. Target: raw next-period return. This was the first walk-forward run on the full 2005-2024 sample.

Result: all three models close to zero Sharpe. R^2 in the GKX ballpark (+0.7% to +0.9% for Lasso and NN against the zero benchmark). XGBoost slightly worse — surprising given the framework calls it the primary model.

Phase 1.5 — add B/M and E/P (Sharadar SF1)

Sourced point-in-time quarterly fundamentals from Sharadar SF1 via Bowen's new `load_fundamentals`. Computed B/M (book equity / market cap) from the ARQ dimension and E/P (TTM net income / market cap) from the ART dimension. Same 120-month sliding training window.

Result: XGBoost's Sharpe explodes from -0.03 to $+0.56$. Lasso barely moves (L1 likely zeroes most of the new coefficients). NN drifts slightly down. R^2 vs zero gets worse for XGBoost ($-0.003 \rightarrow -0.027$) while IC and Sharpe improve — the classic Gu-Kelly-Xiu phenomenon: tree models predict with higher variance once given richer features, so squared-error R^2 punishes them even though rank ordering is better.

Phase 2 — sector-relative target (Layer 2)

Same 7 features. Target changes from raw next-month return to (next-month return) minus (per-(date, sector) mean). This is Layer 2 of the Framework's three-layer sector-neutrality stack (section 3.2).

Result: XGBoost's IC and IC IR improve ($+0.0062 \rightarrow +0.0079$, $+0.090 \rightarrow +0.109$). Drawdowns shrink for XGBoost and NN. But Sharpe drops for all three models — the backtest still uses GLOBAL top/bottom-decile selection, not per-sector top-k. A sector-neutral model feeding a sector-blind portfolio gives up the profitable sector tilts that raw-target models captured.

Decision: keep `target_kind="raw"` as the canonical default. Layer 2 is in the code, ready to switch on the moment Bowen implements Layer 3 (sector-neutral portfolio, `k_per_sector` already exists in `backtest.py` as a warn-only stub).

6. What still needs doing

Phase 3 — XGBoost hyperparameter tuning (next). The current XGBoost is on out-of-the-box defaults (`n_estimators=300`, `max_depth=4`, `lr=0.05`). Tuning on the 2016-2018 validation window via Optuna should give a further Sharpe lift. Will use validation OOS R^2 as the search objective and pin the chosen hyperparameters as the new XGBoostModel defaults.

Phase 4 — Diebold-Mariano model comparison. Frameworks section 8.4 prescribes the adapted DM test for pairwise model ranking. Output: a 3x3 significance table for the final report.

Pending from Bowen — sector-neutral portfolio. Once `k_per_sector` is wired through the backtest loop, re-run Phase 2 and re-evaluate Layer 2's contribution. Should recover the Sharpe drop.

Stretch — lag features and dynamics features. Framework section 3.5. Add `x_{t-1}`, `x_{t-2}` and engineered columns like 3-month momentum change. Skip if time runs out before Week 5 report-writing.

7. Where everything lives

All commits are on the `personb-models` branch on GitHub and pushed to origin. Bowen and Person C can pull at any time.

- Code: `src/factors.py`, `src/models.py`, `src/metrics.py`
- Drivers: `notebooks/personb/01_first_real_backtest.py`, `01b_with_value_factors.py`, `02_sector_relative_target.py`
- Artefacts: `results/01_first_real_backtest/`, `01b_with_value_factors/`, `02_sector_relative_target/`
- Per-phase rationales in `DECISIONS.md` (entries dated 2026-05-22).